

SMART CURRENCY DETECTION AND TRANSACTION VERIFICATION FOR VISUALLY IMPAIRED

PHASE I REPORT

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BONAFIDE CERTIFICATE

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This Project addresses the following Sustainable Development Goal(s):

Goal-10: Reduced Inequality

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ABSTRACT

Cash remains the dominant mode of transaction in cash-driven economies such as India, yet it offers visually impaired users no independent way to confirm the denomination of a currency note. This forces reliance on a sighted companion or on the honesty of the other party, a recurring barrier to financial independence. This project develops an offline assistive system that combines computer vision and text-to-speech to close this gap. A captured note image is passed to a smartphone application, which applies OpenCV preprocessing followed by a YOLOv8-nano object detector trained on eleven classes covering both old and new series Indian currency notes. A confidence-threshold gate decides whether the prediction is reliable enough to announce or whether the note should be captured again; accepted detections are converted to speech by an on-device Piper TTS engine and delivered to the user over Bluetooth audio, with no dependency on internet connectivity at any stage. The detector was trained for 100 epochs on a CPU-only workstation and achieved a precision of 0.980, a recall of 1.000 and a mAP@0.5 of 0.995 on the validation split, with an inference latency of approximately 134 milliseconds per frame.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

YOLO	– You Only Look Once (real-time object detection architecture)
TTS	– Text-to-Speech
mAP	– mean Average Precision
IoU	– Intersection over Union
CNN	– Convolutional Neural Network
FPS	– Frames Per Second
ONNX	– Open Neural Network Exchange
TFLite	– TensorFlow Lite
CPU	– Central Processing Unit
GPU	– Graphics Processing Unit
NPU	– Neural Processing Unit
SDG	– Sustainable Development Goal
dfl	– Distribution Focal Loss

CHAPTER 1

INTRODUCTION

1.1 General

Assistive technology for the visually impaired has traditionally focused on navigation and obstacle avoidance, with comparatively little attention paid to a task that sighted individuals perform without thought several times a day: handling cash. Cash remains a dominant mode of transaction worldwide, and in cash-driven economies such as India it continues to be the default choice for a very large share of everyday purchases. For a visually impaired person, a banknote carries no information that can be perceived by touch alone – Indian currency notes of different denominations are similar in size and texture – and there is presently no independent, on-the-spot way to confirm a note's denomination.

This project, undertaken by Team T032 of the Department of Computer Science and Engineering, Rajalakshmi Engineering College, proposes a camera-based assistive system that recognizes Indian currency denominations and communicates the result to the user through speech. The captured currency frame is processed on a companion smartphone, and the detection and offline text-to-speech pipeline works without any dependency on an internet connection.

1.2 Motivation

In cash-driven economies such as India, the inability to independently verify currency is a recurring, error-prone barrier to financial independence. It forces a visually impaired user to rely on a sighted companion or on the honesty of a shopkeeper or counterpart in the transaction, and the exposure to error is greatest at precisely the point where the mistake would be hardest for the user to catch. Existing assistive currency-recognition tools, reviewed in Chapter 2,

largely identify a note and speak its denomination. This project builds a low-cost, fully offline version of that core capability, using a lightweight detector that runs on a smartphone and a confidence check that avoids announcing uncertain results.

1.3 Objectives

- To design and train a lightweight, real-time object-detection model capable of recognizing Indian currency notes across both old and new note series.
- To preprocess a captured currency frame on the smartphone and run the trained detector on-device, without dependence on any cloud service.
- To deliver the detected denomination to the user as speech through a fully offline text-to-speech engine and Bluetooth audio output, so that the system has no dependency on internet connectivity.
- To incorporate a confidence-threshold recapture mechanism so that uncertain detections are re-checked rather than reported.

1.4 Scope of the Project

This report covers the first phase of the project, which delivers the core recognition capability of the system.

The scope of this phase covers dataset preparation, training and evaluation of a YOLOv8-nano detector, confidence-based acceptance of predictions, and voice announcement of the detected denomination through an offline TTS engine and Bluetooth audio.

Counterfeit or authenticity detection is explicitly excluded from the scope of this project, as is the use of UV/IR imaging hardware and expensive commercial smart-glasses hardware. The system logic is deliberately kept

simple and rule-based – a confidence threshold decides whether a prediction is announced – based on the reasoning that additional complexity increases cognitive load for assistive-technology users.

1.5 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 reviews the existing literature on assistive currency-recognition systems. Chapter 3 analyses the existing systems and states the problem being addressed. Chapter 4 lists the hardware and software requirements. Chapter 5 presents the system architecture, process flow and module description. Chapter 6 describes the methodology followed for dataset preparation, model training and voice feedback. Chapter 7 presents and discusses the implementation results, including the YOLOv8n training outcomes. Chapter 8 concludes the report and outlines directions for further improvement.

CHAPTER 2

LITERATURE SURVEY

2.1 General

This chapter surveys twenty recent works on assistive currency recognition and related vision-based accessibility systems [1]–[20], listed in Table 2.1 in alphabetical order of the first author’s surname. The survey covers detector architectures (CNN [3], [5], [9], [14], YOLOv5 [11], [15], YOLOv8 [7], YOLOv11 [4], YOLO-NAS [2], EfficientDet [1], MobileNet/EfficientNet [18]), feedback mechanisms (TTS and audio [7], [14], [15], [16], [17], tactile [3]), and adjacent problems such as counterfeit detection [2], [6], [10], [12], [13], [19] and multimodal assistive agents [8], [20], in order to position the objectives of this project against the current state of the art.

2.2 Summary of Reviewed Work

Table 2.1 summarises the methodology and key inference of each reviewed work.

Table 2.1 Summary of Literature Survey

No.	Author(s) / Year	Title	Methodology	Inference
1	Bandara (2025) [1]	Money Recognition for the Visually Impaired: A Case Study on Sri Lankan Banknotes	EfficientDet object detection with image preprocessing.	Accurate banknote recognition; establishes a regional benchmark.
2	Chhabra, Goyal (2025) [2]	A Two-Phase Deep Learning Model for Counterfeit Detection of Indian Banknotes using YOLO-NAS and UV Imaging	Two-phase pipeline using YOLO-NAS and UV imaging for security-feature localization.	Detects denomination and counterfeit currency using security-ink features not visible under normal lighting.
3	Edupuganti, Sneha, Vardhan, Kumar (2024) [3]	Generative AI-Based Currency Detector for Visually Impaired	CNN-based image classification with high-resolution camera, tactile buttons, voice commands and audio output.	Recognizes denomination and announces it through audio; tactile controls improve accessibility.
4	Fanzury, Hwang (2026) [4]	Real-Time Identification of Mixed and Partly Covered	YOLOv11 detection with hybrid TFLite / server	Precision 0.951, recall 0.935, mAP@0.5

No.	Author(s) / Year	Title	Methodology	Inference
		Foreign Currency Using YOLOv11 Object Detection	inference; multi-currency and occlusion handling.	0.965; handles mixed and partly covered currencies effectively.
5	Ghanem, Youness, Wahba, Abdelaal (2025) [5]	Recognizing Egyptian Currency for People with Visual Impairment Using Deep Learning Models	CNN-based classification on a regional Egyptian currency dataset.	Provides denomination recognition and addresses a regional gap in accessibility tools.
6	Hantoush (2025) [6]	Artificial Intelligence Model of Iraqi Paper Currency Detection in Vending Machines	CNN classification with embedded deployment and microscopic-level feature analysis.	Combined denomination and authenticity verdict for vending/POS integration.
7	Lalitha, Gayatri, Selvi, Saranya, Ponugoti (2025) [7]	Assisting Visually Challenged Individuals to Distinguish Authentic Currency from Counterfeit	YOLOv8 detection with GTTS speech synthesis; Roboflow annotation and standard detection-metric evaluation.	Provides spoken denomination, currency type and validity in a single audio-based pipeline.
8	Maheswari, Aluvalu, Dhanaraj et al. (2026) [8]	An Assistive Framework for Visually Impaired Using Deep Learning and Natural Language Processing	CNN/RNN/LSTM pipeline with NLP caption generation and speech synthesis.	Narrative spoken scene descriptions including currency identification.
9	Nasayreh, Jaradat, Gharaibeh et al. (2024) [9]	Jordanian Banknote Data Recognition: A CNN-Based Approach with Attention Mechanism	CNN with an attention mechanism and attention-guided feature extraction.	High-accuracy denomination recognition; attention improves feature extraction and robustness.
10	Ovi, Hossain, Biswas (2025) [10]	Quantitative Currency Evaluation in Low-Resource Settings through Pattern Analysis	Lightweight custom CNN with feature-based template matching and a Unified Currency Damage Index.	Identifies denomination, provides a usability/damage score and a counterfeit flag in a compact on-device pipeline.
11	Pasumarthy, Padhy, Yadav, Subramaniam, Rao (2022) [11]	An Indian Currency Recognition Model for Assisting Visually Impaired Individuals	YOLOv5 detection on an annotated, augmented dataset of ~10,000 images (224×224), trained under cluttered, reflective and blurred conditions.	92.71% accuracy on unseen data; robust to reflections, blur and clutter; supported Raspberry Pi and smartphone deployment.
12	Pham, Park, Nguyen, Batchuluun, Park (2020) [12]	Deep Learning-Based Fake-Banknote Detection for the Visually Impaired Using Smartphone Cameras	CNN-based counterfeit detection using RGB image analysis.	Identifies counterfeit banknotes using ordinary smartphone cameras.

No.	Author(s) / Year	Title	Methodology	Inference
13	Pham, Lee, Park, Park (2022) [13]	Deep Learning-Based Fake Multinational Banknotes Detection in a Cross-Dataset Environment	YOLOv3 detection combined with ResNet-18, GoogleNet and Inception-ResNet-v2 fusion.	99.289% mAP for counterfeit classification; strong cross-currency generalization.
14	Puthumalar, Sivaranjani (2024) [14]	Currency Recognition for Visually Impairment People Using Deep Learning Algorithm	CNN with image preprocessing and audio feedback.	Lightweight, real-time denomination recognition designed for accessibility.
15	Sathwik, Balaji, Chandralekha (2026) [15]	Real-Time Currency Identification for Visually Impaired Users Using a Lightweight YOLOv5-Based Object Detection System	YOLOv5-Nano detection with live camera-feed processing and audio feedback.	99.5% mAP at IoU 0.5; low-latency edge inference suitable for real-time assistive applications.
16	Shreya, Siddique, Tasnim (2025) [16]	Real-Time Currency Detection and Voice Feedback for Visually Impaired Individuals	Real-time object detection combined with live voice feedback.	Spoken denomination feedback emphasising real-time responsiveness for daily use.
17	Siddappa, Gowda, Hasham, Kumar (2026) [17]	AI-Based Money Recognition Assistant for Blind People	Lightweight CNN on TensorFlow Lite deployed on an ESP32-CAM with INT8 quantization.	93.8% accuracy, ~1.2 s inference; combines currency recognition with obstacle assistance in an offline wearable system.
18	Subreena, Hossain, Raquib, Murad, Prity, Hanif, Rahimi (2026) [18]	Robust and Real-Time Bangladeshi Currency Recognition: A Dual-Stream MobileNet and EfficientNet Approach	Dual-stream MobileNet and EfficientNet architecture with Explainable AI.	Real-time recognition using an efficient hybrid architecture suitable for edge devices.
19	Wasnik, Temurnikar (2026) [19]	Deep Learning-Based Spectral-Spatial Model for Counterfeit Currency Detection	Dual-branch CNN with attention-guided fusion of RGB and UV/IR imaging.	Robust counterfeit classification under low light, blur and noise.
20	Zhao, Wang, Geng, Yu, Li (2025) [20]	“Less is More”: Reducing Cognitive Load and Task Drift in Real-Time Multimodal Assistive Agents	Multimodal agent design and evaluation with real-time feedback and task-drift analysis.	Design guidelines for reducing cognitive load and task drift in real-time assistive agents.

2.3 Research Gap

Across the reviewed literature, YOLO-family detectors (YOLOv5/v8/v11) consistently achieve high denomination-recognition accuracy [4], [5], [7], [11], [15] – several works report mAP@0.5 above 0.95 [4], [5], [15] – and are

increasingly deployed on lightweight, edge-friendly formats such as TFLite [4], [17], making them well suited to smartphone or microcontroller-adjacent deployment. Text-to-speech and audio feedback are near-universal in this space [3], [7], [14], [15], [16], [17], confirming that voice output is the established modality for communicating detection results to visually impaired users.

However, every system surveyed treats each currency note as an isolated recognition event: the note is identified, its denomination is spoken, and the interaction ends. Additionally, several counterfeit-detection works (rows 2, 12, 13 and 19) [2], [12], [13], [19] rely on UV/IR imaging hardware or cross-dataset fusion models that add cost and computational overhead disproportionate to the everyday goal of denomination recognition; this project deliberately keeps counterfeit detection out of scope in favour of a low-cost, fully offline, rule-based design.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

Current assistive currency-recognition approaches, as surveyed in Chapter 2, fall into three broad categories: (i) CNN or YOLO-based denomination detectors deployed on smartphones or single-board computers such as Raspberry Pi [1], [4], [5], [9], [11], [14], [15], [18]; (ii) GTTS or other cloud-dependent TTS pipelines paired with a detection model [7]; and (iii) spectacle-mounted or wearable prototypes that combine currency recognition with general obstacle detection [17]. These systems are generally able to identify the note currently in view and speak its denomination with reasonably high accuracy [11], [14], [16].

3.2 Limitations of the Existing System

- Several systems rely on cloud-based TTS (e.g., gTTS) [7], which requires an active internet connection and is unsuitable for the fully offline operating requirement of this project.
- Counterfeit-detection variants require additional UV/IR hardware [2], [19], increasing cost and system complexity beyond what is needed for the core denomination-verification use case.
- Most prototypes are demonstrated as standalone recognition tools [6], [10], [16], some with online or computationally heavy components [8], [13], rather than as a lightweight, fully offline pipeline suited to on-phone inference.

3.3 Proposed System

The proposed system addresses these gaps with a lightweight, fully offline design. A currency frame captured by the camera is passed to a smartphone application, which performs OpenCV preprocessing and YOLOv8-nano

inference, applies a confidence threshold to decide whether a detection is reliable, and – if it is – converts the denomination to speech using an offline Piper TTS engine, delivered to the user over Bluetooth audio. If confidence is below the threshold, the system does not announce the result and asks for the note to be captured again instead of reporting an uncertain result.

3.4 Problem Statement

Visually impaired users have no independent way to confirm a currency note's denomination. In cash-driven economies such as India, this is a recurring, error-prone barrier at exactly the point where mistakes are hardest for a visually impaired user to catch. This project aims to close the recognition part of that gap with a low-cost, fully offline system that performs real-time denomination detection and announces the result by voice.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 Hardware Requirements

The prototype is intentionally designed around hardware that is already commonly available: a workstation for training the model, a smartphone for on-device inference, and Bluetooth earphones or a speaker for audio output. Table 4.1 lists the hardware components.

Table 4.1 Hardware Requirements and Approximate Cost

Component	Use	Approx. Cost
Training workstation (12th Gen Intel Core i7-1255U, CPU only)	Dataset preparation, model training and evaluation	Already available
Smartphone	Local inference (OpenCV, YOLO, TTS)	Already owned
Bluetooth earphones / speaker	Audio feedback to the user	Already owned

4.2 Software Requirements

Table 4.2 lists the software and technology stack used in this work. Python environment and dependency management is handled exclusively through uv, with pyproject.toml as the single source of truth and uv.lock committed for reproducibility. Docker is used strictly to standardize the training/evaluation environment across the team and is not used on the phone-side inference app, which cannot run containers.

Table 4.2 Software / Technology Stack

Stage	Tool / Technology	Purpose
Image input	Captured currency frame (image)	Input to the smartphone application
Preprocessing	Python 3.11 + OpenCV	Resize, crop, normalize, denoise, brightness enhancement
Detection	YOLOv8-nano (Ultralytics)	Currency denomination detection
Training	Python 3.11 + Ultralytics / PyTorch	Model training and validation

Stage	Tool / Technology	Purpose
Environment / dependency mgmt.	uv (pyproject.toml + uv.lock)	Reproducible Python environment
Dev / training containerization	Docker + docker-compose	Standardized training / evaluation environment only
Confidence logic	Python rule-based threshold logic	Accept reliable predictions / request recapture
Voice output	Piper TTS (offline)	Convert detected denomination to speech
Audio delivery	Bluetooth earphones / speaker	Deliver spoken result to the user
Code quality	ruff format, ruff check, type hints	Formatting, linting and static typing

CHAPTER 5

SYSTEM DESIGN

5.1 System Architecture

Figure 5.1 shows the overall architecture of the system. A captured currency frame is passed to the smartphone application, whose processing system contains two blocks: Currency Detection (image preprocessing, YOLO detection of the denomination, and confidence scoring using the quantized model) and Voice Feedback (label mapping of the denomination to text, on-device text-to-speech, and generation of a user-friendly voice message). The generated audio is delivered to the output unit, where Bluetooth earphones or a speaker announce the denomination, for example “Rs. 500”, to the user.

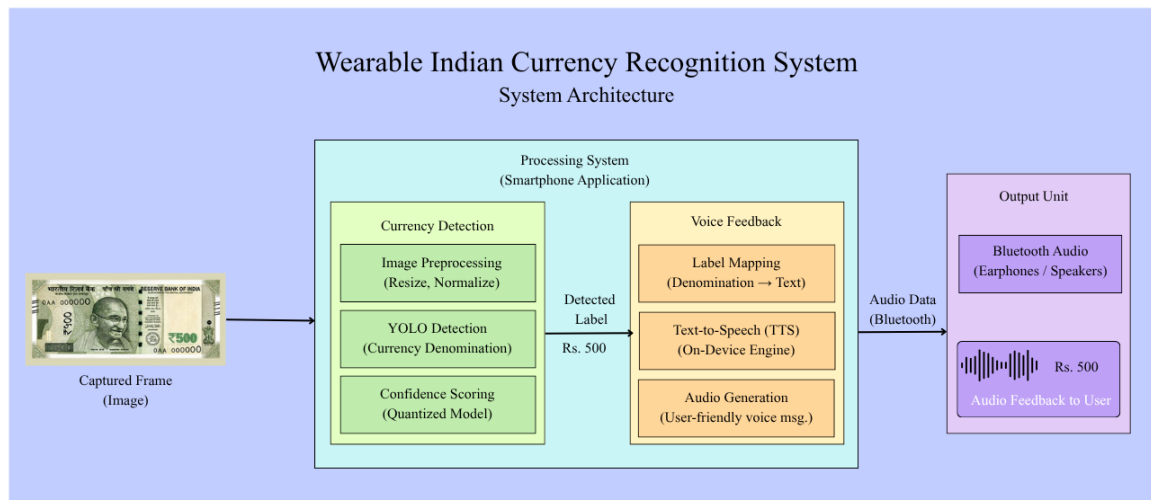


Figure 5.1 System Architecture – Wearable Indian Currency Recognition System

5.2 Process Flow

The process flow of the system is as follows:

1. Image input – a camera captures the currency note and the frame is provided to the smartphone application.
2. Preprocessing – OpenCV resizes and normalizes the image and applies basic brightness enhancement or denoising.

3. Denomination detection – the YOLOv8n model predicts the currency class, bounding box and confidence score.
4. Confidence handling – a confidence threshold determines whether the prediction is reliable; a low-confidence result triggers a request to capture the note again.
5. Voice output – the denomination label is mapped to text and the Piper TTS engine converts it into speech.
6. User feedback – the result is delivered through Bluetooth earphones or a speaker.

5.3 Module Description

The system is organized into two modules, summarised in Table 5.1.

Table 5.1 Module Description

Module	Function	Key Components
M1 – Currency Detection	YOLO-based denomination detection from a captured frame	OpenCV preprocessing, trained/quantized YOLO model, confidence scoring
M2 – Voice Feedback	TTS output announcing the detected denomination	Label-to-speech mapping, on-device Piper TTS engine, Bluetooth audio output

Inference runs entirely on the connected mobile phone, and no component of the pipeline requires an internet connection.

5.4 Project Structure

The codebase follows a modular structure in which the detection and audio modules are independently runnable and demoable. Scripts only orchestrate training, evaluation and deployment entry points, while all core logic resides in dedicated source modules for detection, audio, configuration and shared utilities. Model hyperparameters and thresholds are centralized in a

configuration module, and the Python environment is managed with uv for reproducibility.

CHAPTER 6

METHODOLOGY

6.1 Dataset Preparation

The detector was trained on an annotated Indian currency note dataset covering eleven classes spanning both the old and new note series (e.g., ₹10, ₹20, ₹50, ₹100, ₹200, ₹500 and ₹2000, each tagged by series where applicable). The dataset was split into 459 training images and 23 validation images, formatted for YOLO training with a data.yaml configuration file. Class mappings and split paths were verified as the first step before training began.

6.2 Model Selection

YOLOv8-nano was selected as the detection backbone specifically for its low latency on phone-side inference: at roughly 3 million parameters and 8.2 GFLOPs, it is small enough to run efficiently on a mid-range smartphone CPU/NPU while still supporting transfer learning from COCO-pretrained weights. All model hyperparameters – image size, confidence threshold, IoU threshold and the class list – are centralized in a dedicated configuration module rather than hardcoded inline, so that they can be re-tuned from the actual training results without touching the detection or audio code.

6.3 Training Configuration

The model was trained for 100 epochs on a CPU-only workstation (12th Gen Intel Core i7-1255U, no GPU), using the Ultralytics YOLOv8 training pipeline with an image size of 640×640 and a batch size of 16. Table 6.1 summarises the key training configuration parameters.

Table 6.1 YOLOv8n Training Configuration

Parameter	Value
Base model	YOLOv8n (nano)
Framework	Ultralytics 8.4.126, PyTorch 2.13.0 (CPU build)
Hardware used for training	12th Gen Intel Core i7-1255U (CPU only, no GPU)
Python version	3.11.13
Number of classes	11 (old and new series Indian currency notes)
Training images	459
Validation images	23
Image size	640 × 640
Batch size	16
Epochs	100
Optimizer	AdamW (auto-selected), lr0 = 0.000667, momentum = 0.9
Early-stopping patience	20 epochs (never triggered)
Augmentation	HSV jitter, horizontal flip (p = 0.5), mosaic, random erasing
Total training time	5.147 hours
Model size	130 layers, 3,012,993 parameters, 8.2 GFLOPs

6.4 Confidence-Threshold Recapture Logic

Rather than reporting every detection, the smartphone application applies a confidence threshold (tuned from the training results and stored in the configuration module) to decide whether a prediction should be spoken. If the top detection's confidence is below the threshold, the system asks the user to capture the note again instead of announcing an uncertain denomination – this avoids the failure mode of confidently mis-stating a note's value to a user who cannot visually double-check the result.

6.5 Offline Voice Feedback

Text-to-speech is implemented using Piper TTS, chosen because it is fully offline, lightweight and provides good voice quality through a simple Python API, in line with the project's strict offline-only constraint. Cloud or API-

based TTS engines such as gTTS were explicitly avoided because they require an active internet connection, which the system is designed never to depend on.

6.6 Offline Operation

The complete pipeline – preprocessing, inference, confidence handling and speech synthesis – runs on the smartphone without any network connection. No tool, API or library that assumes internet access is used anywhere in the pipeline, including the text-to-speech engine, so the system behaves identically with or without connectivity.

CHAPTER 7

IMPLEMENTATION AND RESULT DISCUSSION

The YOLOv8n detector was trained for the full 100 epochs on a CPU-only workstation; the results below are drawn directly from the training run's console log and per-epoch validation metrics. Figure 7.1 plots the training/validation loss components and the precision, recall and mAP metrics across all 100 epochs, discussed section by section below.

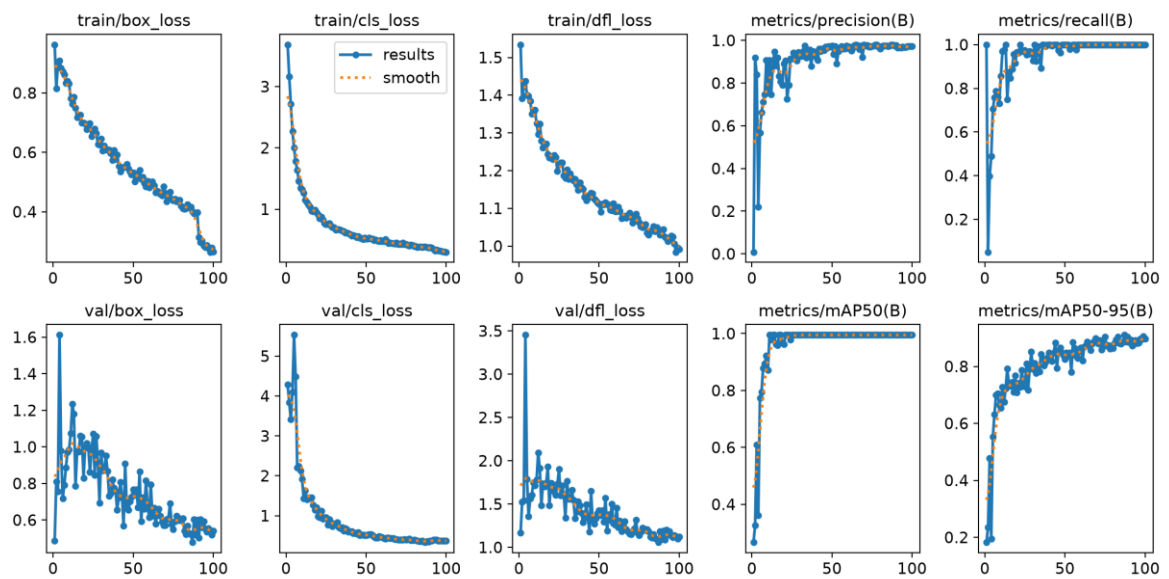


Figure 7.1 Training and Validation Loss and Metric Curves over 100 Epochs

7.1 Training Loss Trend

Table 7.1 Training Loss Trend (Epoch 1 / 50 / 100)

Loss	Epoch 1	Epoch 50	Epoch 100	Trend
box_loss	1.252	0.531	0.266	Steady decline – model learns to localize bounding boxes
cls_loss	3.914	0.520	0.302	Steep early drop, gradual convergence – class separation improves
dfl_loss	1.724	1.112	0.991	Slow, steady – distribution focal loss refines box edges

box_loss fell 79% across training, reflecting steadily improving bounding-box localization; cls_loss fell 92%, the dominant early loss as the model initially

struggled to separate 11 visually similar currency classes before learning quickly; dfl_loss declined more gradually (43%), consistent with its role in refining fine-grained box boundaries.

7.2 Precision and Recall Progression

Table 7.2 Precision Progression

Phase	Value	Interpretation
Epoch 1	0.008	Near-zero – model predicts everything, most predictions wrong
Epoch 2	0.918	Sharp jump – model learns to be selective
Epoch 3–10	0.22 – 0.91	Volatile – model oscillates between cautious and aggressive
Epoch 11+	0.88 – 0.98	Stable and high – model is confident and correct
Best (epoch 87)	0.980	98% of detected objects are actual currency notes

Table 7.3 Recall Progression

Phase	Value	Interpretation
Epoch 1	1.000	Detects everything (but mostly wrong)
Epoch 2	0.050	Collapses – model becomes overly conservative
Epoch 3–10	0.40 – 0.86	Recovers steadily
Epoch 11+	0.93 – 1.00	Catches nearly all currency notes
Best (epoch 87)	1.000	Finds every currency note in the validation set

The epoch-2 recall collapse coincides with the sharp precision jump the same epoch – the model swung from “predict everything” to “predict almost nothing” before settling into a stable balance from epoch 11 onward.

7.3 mAP Progression

Table 7.4 mAP@0.5 Progression

Phase	Value	Interpretation
Epoch 1	0.266	Low – model barely overlaps ground truth
Epoch 3	0.610	Rapid improvement
Epoch 7	0.878	Crosses into production-grade territory
Epoch 11	0.995	Ceiling reached

Phase	Value	Interpretation
Epochs 12–100	0.962 – 0.995	Stays at or near ceiling for the remaining 89 epochs

Table 7.5 mAP@0.5:0.95 Progression

Phase	Value	Interpretation
Epoch 1	0.183	Very loose detections
Epoch 11	0.727	Good, but boxes not pixel-precise
Epoch 44	0.883	Boxes getting tighter
Epoch 75	0.905	Near-optimal box precision
Best (epoch 87)	0.910	91% overlap accuracy at strict IoU thresholds

mAP@0.5 reached its 0.995 ceiling by epoch 11 and stayed there for the remaining 89 epochs, while the stricter mAP@0.5:0.95 metric kept improving all the way to epoch 87 – evidence that the model continued refining bounding-box precision on objects it had already learned to detect, rather than merely memorizing the training set.

7.4 Per-Class Detection Performance

All eleven classes reached mAP@0.5 = 0.995; the variance across classes shows up only in the stricter mAP@0.5:0.95 metric, summarised in Table 7.6.

Table 7.6 Per-Class Detection Performance (mAP@0.5:0.95)

Class	mAP@0.5:0.95	Notes
₹50 (New)	0.995	Best – single instance, model nailed it
₹10 (Old)	0.951	Strong – distinct visual features
₹20 (Old)	0.945	Strong
₹2000	0.895	Average
₹200	0.895	Average
₹20 (New)	0.895	Average
₹500 (New)	0.895	Average
₹50 (Old)	0.895	Average
₹100 (New)	0.889	Slightly below average
₹100 (Old)	0.846	Lowest – only 2 validation images, likely harder variations

No class performs significantly worse than the others; the spread (0.846–0.995) is driven mainly by the small number of validation images available per class (1–4 images), rather than by any structural weakness in the model.

Figure 7.2 presents the confusion matrix computed on the validation split. Every predicted class lies on the diagonal, matching its true class with no off-diagonal confusions anywhere in the matrix, corroborating the precision and recall values reported above and confirming that the detector does not confuse visually similar denominations, such as the old- and new-series notes of the same face value.

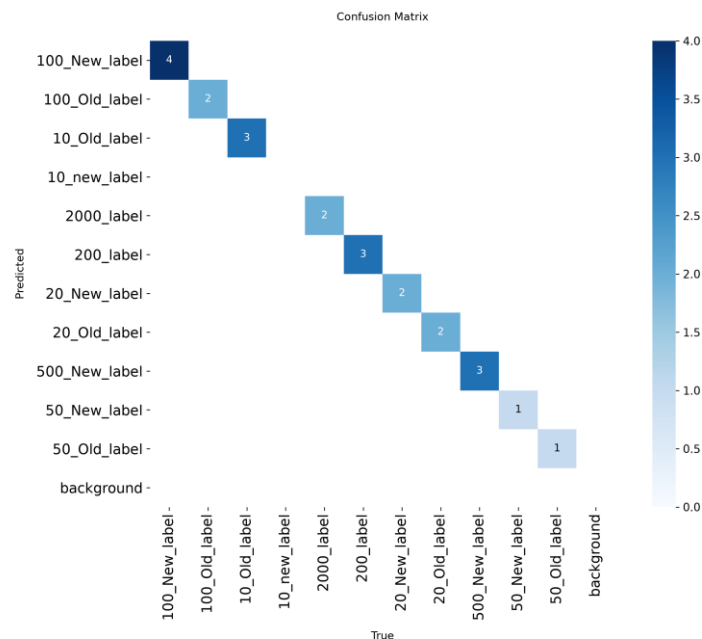


Figure 7.2 Confusion Matrix on the Validation Set

Figures 7.3 and 7.4 show a representative validation batch with the ground-truth annotations and the corresponding model predictions overlaid on the same images. The predicted class labels and confidence scores closely match the ground-truth labels across all sixteen sample notes in the batch, including notes captured at an angle, partially folded, or photographed against varied backgrounds and lighting.



Figure 7.3 Ground-Truth Labels – Validation Batch Sample



Figure 7.4 Model Predictions – Validation Batch Sample

7.5 Overfitting Assessment

No significant overfitting was observed. Validation losses for all three loss components decreased steadily throughout training with no point at which validation loss rose while training loss continued to fall – the classic overfitting signature. $mAP@0.5$ plateaued at its ceiling from epoch 11 without ever dropping sharply and recovering, which would suggest memorization, and $mAP@0.5:0.95$ continued climbing for 76 further epochs, indicating genuine, ongoing refinement of box precision. YOLOv8n's small capacity (about 3 million parameters) relative to the 459-image training set further limits the model's ability to memorize the data even if it were inclined to. Early stopping (patience = 20) never triggered, since the model kept matching or beating its best $mAP@0.5$ well past epoch 80 – the ideal outcome of a model

that genuinely needed the full 100 epochs to converge rather than stalling early.

7.6 Inference Latency

Table 7.7 Inference Latency Breakdown (CPU, no GPU)

Stage	Latency
Preprocess	1.7 ms
Inference	131.0 ms
Postprocess	1.4 ms
Total (CPU)	~134 ms (~7.5 FPS)

An end-to-end latency of approximately 134 ms per frame (about 7.5 FPS) on a CPU-only workstation is acceptable for an assistive tool that announces a result once per capture rather than requiring continuous video-rate throughput. On a phone with GPU or NPU acceleration – the actual target deployment – this figure is expected to drop substantially, to roughly 20–40 ms per frame.

7.7 Comparison with Literature

The achieved validation performance (precision 0.980, recall 1.000, mAP@0.5 0.995) compares favourably with the strongest results reported in the literature survey of Chapter 2, several of which report mAP@0.5 in the 0.95–0.965 range using comparable YOLO-family detectors. This suggests that the YOLOv8n configuration and training regime used here are competitive with the current state of the art for Indian-currency denomination detection, while remaining a substantially smaller and faster model than the CNN-attention or dual-branch architectures used in some of the surveyed counterfeit-detection systems.

7.8 Current Development Status

Table 7.8 summarises progress against the development checklist at the time of this report.

Table 7.8 Development Status Checklist

No.	Item	Status
1	Dataset preparation	Done
2	Train YOLOv8n	Done
3	Export model (ONNX / TFLite)	Done
4	Detection module	Done
5	Config tuning from training results	Done
6	Audio module (Piper TTS)	Done
7	Detection–audio pipeline integration	Done
8	End-to-end test (captured frame → detect → speak)	Done

The dataset, trained detector, exported model, detection module, tuned configuration and audio module are complete. Integration of the detection and audio modules into a single pipeline, unit tests, the end-to-end test and documentation remain to be completed before this phase is considered finished.

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

This report has presented the design, methodology and results of a fully offline assistive system for Indian currency denomination detection aimed at visually impaired users. A YOLOv8-nano detector trained on eleven currency classes achieved a validation precision of 0.980, a recall of 1.000 and a mAP@0.5 of 0.995 with no evidence of overfitting, and an end-to-end inference latency of approximately 134 ms on CPU-only hardware, competitive with the strongest results reported in the literature survey. The detection and offline text-to-speech modules are complete; pipeline integration, automated testing, the end-to-end demonstration and documentation remain as the immediate next steps.

8.2 Future Work

Future work will focus on strengthening the current system. The dataset will be enlarged with more images per class and a wider range of lighting, blur, occlusion and reflective-surface conditions, so that per-class performance can be evaluated on a larger validation set. Inference latency and power usage will be benchmarked on the actual target smartphone hardware, and the detection and voice modules will be packaged into a single mobile application. The robustness of the detector under low light, blur and partial occlusion will also be evaluated systematically.

Consistent with the project's design philosophy, the system will be kept deliberately simple and rule-based – a confidence threshold rather than a complex decision framework – based on the reasoning that added complexity increases cognitive load for assistive-technology users. Counterfeit or authenticity detection, UV/IR imaging hardware and expensive commercial

smart-glasses hardware will continue to be explicitly excluded from the project's scope.

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